



Integration and System Testing Report (Part 1)

Part 1: Integration and System Testing of the MobileFoodDeliveryApp

This report documents the application of integration and system testing methods, following best practices, for the MobileFoodDeliveryApp codebase.

1. Project & Team Information

Field	Value
Assignment Name	Part 1: Integration and System Testing of the MobileFoodDeliveryApp
Course	Software Testing (Autumn 2025)
Group Members	Muditha Kumara (muditha.kumara@centria.fi), Chuks Henry
Date Submitted	15/10/2025
App Version/Commit	Alpha 1.0
Code Repository	https://github.com/Muditha-Kumara/SoftwareTesting/tree/2.1.phase1

2. Roles & Responsibilities

Member	Role
Muditha Kumara	Test Lead, Integration Specialist
Chuks Henry	System Tester

3. Integration Testing

3.1. Strategy Chosen

Approach: Top-Down Integration Testing

Rationale: This approach allows us to validate the main application flow early, using stubs for unfinished lower-level modules (e.g., Payment and Notification services). It helps catch integration issues at the user interface and controller level before moving to backend details.

3.2. Modules & Sequence

Sequence	Module	Stub/Driver Used
1	Application Controller (main.py)	Stubs for PaymentProcessing, NotificationService
2	Order Placement (Order_Placement.py)	Real module

Sequence	Module	Stub/Driver Used
3	Payment Processing (Payment_Processing.py)	Stubbed responses

3.3. Test Scenarios & Results

Test Case	Description	Expected Result	Actual Result	Pass/Fail
TC1	Place order with valid cart and payment	Order confirmed, payment processed	Success, order ID returned	Pass
TC2	Place order with invalid payment (stubbed failure)	Payment declined, order not confirmed	Failure message shown	Pass
TC3	Place order with unavailable menu item	Error message, order not placed	Error shown	Pass

3.4. Key Findings

- Top-down integration exposed UI-to-backend communication issues early.
- Stubbing PaymentProcessing allowed simulation of both success and failure cases.
- Error handling for unavailable items and payment failures worked as expected.

4. System Testing

Each member performed one functional and one non-functional test on a selected module.

4.1. Functional Tests

Member	Module	Test Case	Acceptance Criteria	Result
Muditha Kumara	Payment_Processing.py	Valid credit card processes payment	Payment succeeds, transaction ID returned	Pass
Chuks Henry	Order_Placement.py	Add item to cart and place order	Item added, order confirmed	Pass

4.2. Non-Functional Tests

Member	Aspect	Test Type	Metric/Tool	Criteria	Result
Muditha Kumara	Performance	Simulate 50 concurrent orders	Python threading, response time	<2s per order	Pass
Chuks Henry	Usability	Heuristic evaluation of checkout	User feedback, checklist	No critical usability issues	Pass

5. Reflections & Lessons Learned

5.1. Challenges

- Creating realistic stubs for PaymentProcessing required careful design.
- Ensuring test isolation between integration and system tests.

5.2. Observations

- The top-down approach helped catch integration bugs early.
- System tests confirmed both functional correctness and non-functional quality.

5.3. Lessons Learned

- Integration testing is essential for validating module interactions, not just individual correctness.
- Non-functional testing (performance, usability) is critical for real-world readiness.

6. Attachments & Evidence

- Screenshots of test runs and coverage reports (see attached images).
- Sample output logs from performance and usability tests.

End of Report